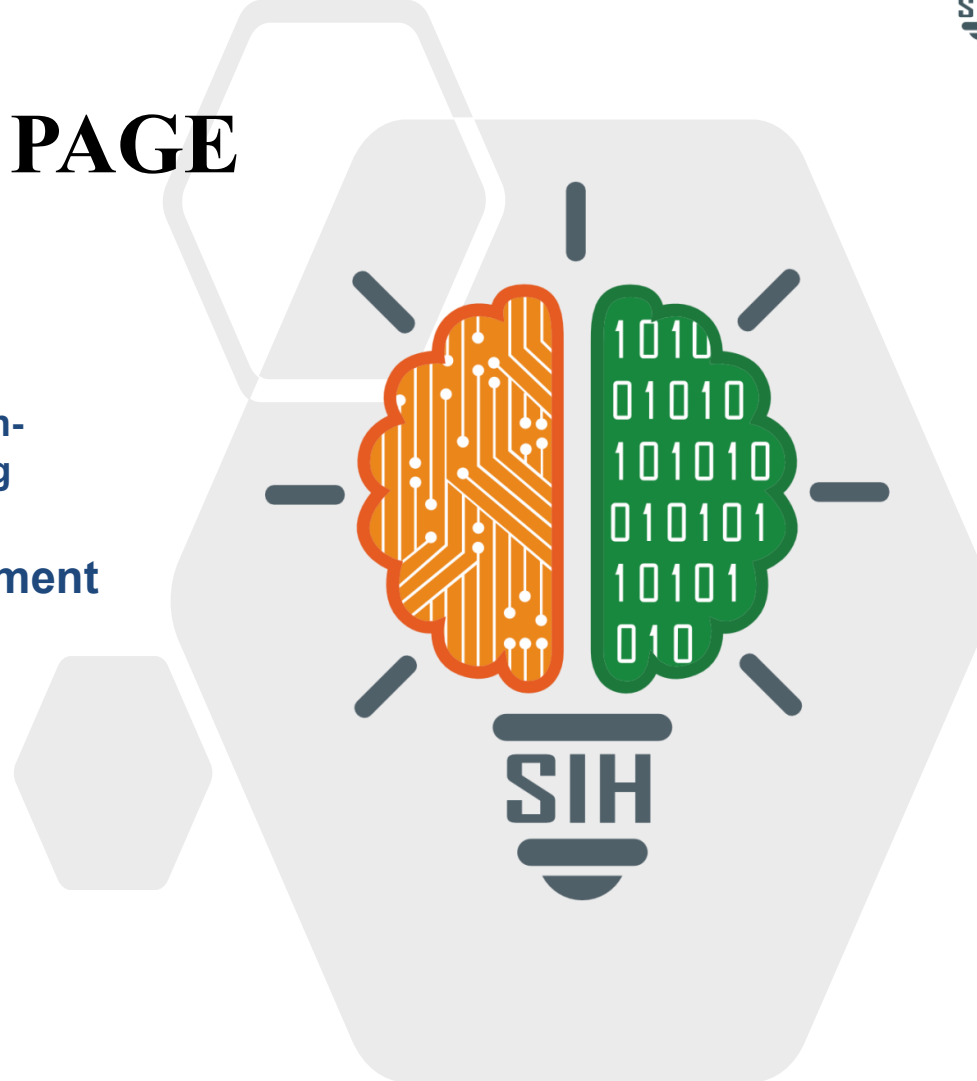


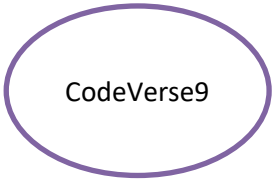
SMART INDIA HACKATHON 2026



TITLE PAGE

- Problem Statement ID – **SIH26021**
- Problem Statement Title- **Honey Chain: A block chain-based system for honey traceability and smart beekeeping management.**
- Theme- **Agriculture, FoodTech & Rural Development**
- PS Category- **Software**
- Team ID- **133847**
- Team Name (Registered on portal)- **CodeVerse9**
- Live Prototype- honey-chain-rust.vercel.app
- Demo Video- youtu.be/-nWbjNHHzBw





HONEY CHAIN: Blockchain Honey Traceability + AI-IoT Smart Beekeeping

❖ Proposed Solution (Describe your Idea/Solution/Prototype)

Honey Chain gives every hive, harvest batch and jar a tamper-proof digital identity on a permissioned blockchain. Consumers verify authenticity by scanning a QR code; IoT + AI keep colonies healthy and productive.

• Detailed explanation of the proposed solution

1. **Register** – each KVIC bee box gets a Hive ID (owner, village, GPS, cluster, flora) written to the ledger.
2. **Sense** – ESP32 hive node streams weight, brood temperature, humidity, colony sound, CO₂ and entrance activity; edge AI diagnoses varroa, foulbrood, queenless, pre-swarm and starvation and forecasts 30-day yield.
3. **Anchor** – 14-day telemetry is hashed into a Merkle root and sealed on-chain; raw data stays local, so a 2G link is enough.
4. **Harvest** – beekeeper picks hives, enters kg + flora → batch ID + printable QR; yield-per-hive plausibility is checked automatically.
5. **Certify** – NABL lab records moisture, HMF, sucrose, reducing sugars, ash, C4 sugar, NMR profile and rice-syrup marker (SMR); pass/fail is computed from limits and cannot be edited.
6. **Move** – processor / distributor / retailer handovers and retail packing are logged; every write prints a signed receipt.
7. **Verify** – consumer scans the QR → trust score 0-100, origin, lab table, journey and validator signatures; forged codes return "no such batch".

• Innovation and uniqueness of the solution

- Lab-agnostic trust layer: anchors NMR, SMR, C4 and HMF results, so syrups engineered to pass basic tests are still rejected (CSE 2020)
- Yield plausibility: claimed kg vs hive-weight telemetry exposes blending ("95 kg from 1 hive" is flagged)
- Telemetry anchored by Merkle root: auditable claims with minimal bandwidth and cost
- Permissioned PoA (KVIC / FSSAI / lab validators): no mining, no gas, accountable writers
- Explainable AI with advice in English + Hindi; offline-first installable PWA built for villages

• How it addresses the problem

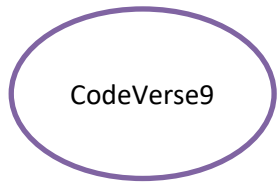
Problem (PS SIH26021)	Honey Chain answer
Counterfeit honey	Lab results incl. NMR/SMR sealed per batch; failures are permanent; QR verifies each jar
Low consumer trust	Scan-to-verify trust score, origin and journey read from the ledger, not from the seller
Weak market linkage	Verified batches + custody trail support premium pricing and direct/retail buyers
No traceability or hive-management support	Hive → harvest → jar trail; IoT-AI alerts, advice and yield forecast

• Expected solution coverage (per problem statement)

- ✓ **Blockchain prototype + QR consumer authentication** – built and live
- ✓ **IoT hive monitoring + AI analytics** – built (simulated sensors, explainable model)
- ✓ **Scalable KVIC deployment framework** – phased plan, costs, risks (slide 4)

• Trust score shown to the buyer (0-100)

+25 hives registered · +15 harvest logged · +40 lab PASS · +10 telemetry anchored · +10 custody / retail trail · –20 implausible yield per hive
Lab FAIL or score < 50 → REJECT. Unknown / forged QR → "no such batch". Ledger integrity is re-checked on every load.



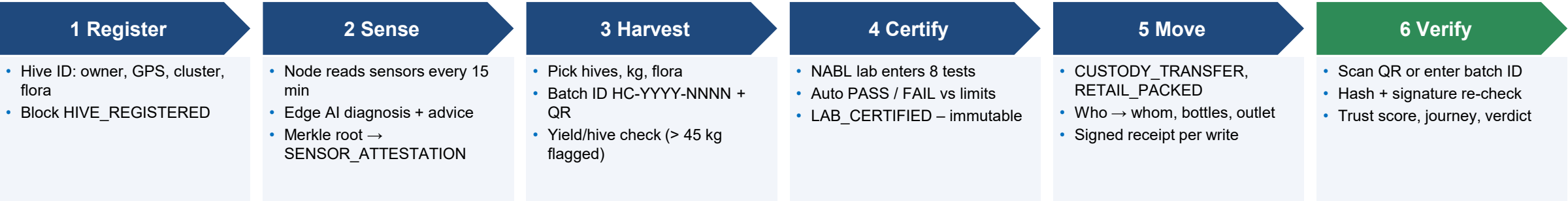
TECHNICAL APPROACH



❖ Technologies to be used (e.g. programming languages, frameworks, hardware)

Layer	Technologies	Role
Frontend	React 18, Vite, JavaScript; installable offline-first PWA (service worker); html5-qrcode; EN / हिन्दी	Consumer, beekeeper, lab, supply-chain and officer portals
Ledger	SHA-256 hash-chain (Web Crypto); Proof-of-Authority round-robin validators; Merkle trees	Tamper-evidence; integrity check on every load
Smart contract	Solidity 0.8 (roles, batch, certify, custody) → Hyperledger Besu / Fabric-class network	Production ledger; a lab FAIL is final
IoT hardware	ESP32, HX711 + load cell, DS18B20, DHT22, MEMS mic (I ² S), CO ₂ sensor, LoRa, solar + Li-ion	15-min readings; 1 gateway per 20-40 hives
AI / analytics	Feature model: brood-temp variance, sound shift, weight slope, activity drop, CO ₂ → softmax; CNN (TFLite Micro)	5 conditions + healthy; 30-day yield forecast (80% range)
Data	On-chain: hashes only. Off-chain: time-series store; IPFS for lab PDFs	Privacy, low cost
DevOps / security	GitHub Actions CI; Pages / Vercel / Docker + nginx; HTTPS; role-based keys (Ed25519 planned)	Repeatable, secure deployment

❖ Methodology and process for implementation (Flow Charts/Images/ working prototype)



Design rule: only hashes and attestations go on-chain; raw sensor data stays local and is proven later via the Merkle root. **AI:** softmax over 6 states, 24 / 24 injected conditions identified in tests. **Data:** synthetic records modelled on real districts, flora and public parameters (slide 6).

How we use AI (beyond the blockchain)

[Built] Hive-health model: sensor features → varroa, foulbrood, queenless, pre-swarm, starvation (24/24 on simulated tests)
[Built] 30-day yield forecast (80% range) + fraud flag when claimed kg per hive exceeds what hive data supports
[Plan] CNN on hive audio via TFLite Micro, trained on open datasets (refs 8, 9); advice shown with confidence
[Plan] Comb-photo disease check on the phone; Hindi voice entry and advice for low-literacy users

How we use Cloud

[Today] PWA served from a CDN (GitHub Pages / Vercel); CI/CD with GitHub Actions
[Plan] LoRaWAN → MQTT broker → time-series DB; Merkle root anchored on-chain
[Plan] Validator nodes + indexer as containers on MeitY-empanelled / GI Cloud; KMS/HSM keys
[Plan] Cloud GPU training, OTA model updates, drift monitoring; object storage + IPFS; retailer API; SMS / WhatsApp alerts

Analysis of the feasibility of the idea

- **Technical:** working prototype live (ledger, QR verify, lab portal with NMR/SMR, hive AI, receipts, PWA); mature open-source stack
- **Hardware:** ESP32 node ≈ ₹1.5-2.5k per hive*; one LoRa gateway serves 20-40 hives within ~5 km
- **Economic:** near-zero ledger cost (PoA, no gas); QR label ≈ ₹0.1-0.3 per jar*
- **Operational:** fits NBHM / KVIC clusters, NABL labs and FSSAI oversight; no new institution needed
- **Regulatory:** minimal personal data (name, village); limits FSSAI/BIS-style, final limits set by the notified lab
- **Scalable:** validator nodes per KVIC region; hashes on-chain, raw data off-chain

Potential challenges and risks

- **Garbage-in:** a dishonest lab or beekeeper can enter false data
- **Adulteration that beats basic tests:** engineered syrups pass C4/HMF (CSE 2020)
- **Connectivity and literacy:** patchy 2G/4G, low digital skills
- **QR cloning:** copying a genuine label onto fake jars, refilling
- **Field hardware:** sensor drift, theft, power, weather
- **Adoption and governance:** cluster buy-in, cost sharing, who runs validators and keys

Strategies for overcoming these challenges

- **Trust in inputs:** accredited-lab keys, random re-tests, IoT weight cross-check vs claimed yield
- **Lab-agnostic:** anchor NMR + SMR alongside C4/HMF; new tests plug in
- **Village-ready:** offline-first PWA, EN/HI, QR-first flows, assisted entry via KVIC field officers
- **Anti-cloning:** per-jar serials + single-use claim codes, tamper seals (roadmap)
- **Hardware:** calibration, solar, IP-rated enclosure, sensor-anomaly flags
- **Governance:** multi-stakeholder validators (KVIC, FSSAI, labs); hardware key storage; pilot via SHG trainers

Unit economics – planning estimates* (to be validated in the pilot)

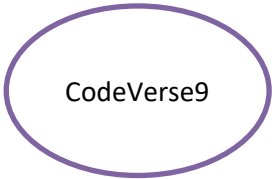
Item	Estimate	Note
Hive node (ESP32 + sensors)	₹1.5-2.5k / hive	solar + LoRa
LoRa gateway (per cluster)	₹15-25k	20-40 hives each
QR label per jar	₹0.1-0.3	printed by beekeeper
Pilot: 100 hives, 3-5 gateways	≈ ₹2-3.8 lakh	hardware only

Phased rollout and validation plan



- Validation first: 10-minute interviews with 5 beekeepers, 2 NABL labs, 1 KVIC field officer (script prepared)
- Pilot KPIs: % batches with QR + lab certificate, scans/month, price realisation, colony-loss rate

*Planning estimates, not quotes.



IMPACT AND BENEFITS



❖ Potential impact on the target audience

Audience	Today	With Honey Chain
Rural beekeepers (KVIC Honey Mission)	Cannot prove purity; sell to middlemen at commodity prices; late disease detection	Verifiable quality → direct buyers and premium price; early alerts and yield forecast reduce colony loss
Consumers	Cannot tell real honey from syrup blends; labels are easy to copy	Scan-to-verify in seconds: origin, lab report, journey, trust score; fake batches flagged before purchase
KVIC / FSSAI / regulators	No live view of production or adulteration; manual audits	Cluster dashboards, tamper-evident audit trail, adulteration flags, hotspot targeting for extension
Processors, retailers, e-commerce	Brand and recall risk from unverifiable supply	Trusted supply chain with custody trail; verified, premium-ready listings
Labs and exporters	Reports on paper, easy to dispute or re-use	Signed, immutable certificates (NMR/SMR/C4/HMF) reusable as export evidence

❖ Benefits of the solution (social, economic, environmental, etc.)

Social	Economic	Environmental	Governance
- Trust in rural produce - Livelihood security incl. SHGs / women beekeepers - Bilingual, low-literacy design	- Higher realisation per kg - Less fraud loss for buyers - Export-ready traceability - Lower colony-loss cost	- Healthier colonies → pollination - Targeted treatment, less chemical - Low-energy ledger (no mining) - Paperless certificates	- Tamper-evident audit trail - Data-driven KVIC planning - Aligned to SDG 1, 2, 8, 12, 15

WHY NOW – public evidence

17 of 22

honey samples tested were adulterated with sugar syrup (77%)
CSE investigation, Dec 2020

46%

of imported honey in the EU was suspected of adulteration in 2022 (EU coordinated action)
via Frontiers Sust. Food Syst., 2025

1.4 lakh MT

natural honey produced in India in 2024; ~1.07 lakh MT exported in FY 2023-24
PIB - National Beekeeping & Honey Mission

₹500 crore

NBHM outlay, FY 2020-21 to 2025-26: production is growing, trust must keep pace
PIB

Pilot KPIs (baseline measured at pilot start)

- 100% of pilot batches with QR + lab certificate
- Scans / month; adulterated batches flagged per 1,000
- Price realisation vs unlabelled honey; colony-loss rate

❖ Details / Links of the reference and research work

Market, regulatory and standards evidence

1. **CSE – "Laboratory results of honey testing" (Dec 2020): NMR tests on branded honey**
cdn.downtoearth.org.in/report_laboratory_results_of_honey_testing_20201201.pdf
2. **FSSAI press release reviewing the CSE honey-sample issue (3 Dec 2020)**
fssai.gov.in/upload/press_release/2020/12/5fc8eb6d70965Press_Release_Honey_Sample_03_12_2020.pdf
3. **PIB – National Beekeeping & Honey Mission: outlay, production, exports**
pib.gov.in/PressReleasePage.aspx?PRID=2185400 | National Bee Board: nbb.gov.in
4. **The Tribune – "Honeytrap": syrups designed to bypass purity tests**
tribuneindia.com/news/nation/honeytrap-cse-investigations-claim-nefarious-adulteration-business-of-honey-designed-to-bypass-purity-tests-178930
5. **Business Standard – FSSAI on the SMR (rice-syrup marker) test**
business-standard.com/article/companies/fssai-questions-cse-for-not-conducting-smr-test-on-honey-brands-120120301199_1.html
6. **Codex Alimentarius – Standard for Honey, CXS 12-1981 (rev. 2001; amended 2019, 2022)**
fao.org/input/download/standards/310/cxs_012e.pdf | Indian standard: BIS IS 4941 (honey)

Data used in the prototype (transparency)

- **Real public data:** CSE 2020 test findings, PIB / NBHM production and outlay figures, FSSAI / BIS / Codex honey parameters (refs 1-6) - used for the problem evidence, lab limits and impact statistics.
- **Prototype dataset:** 15 hives, 10 batches and 51 ledger blocks across 8 real districts in 6 states (Bharatpur, Muzaffarpur, Ludhiana, South 24 Parganas, Saharanpur, Nashik, Nagpur, Sawai Madhopur) using real flowering regions. Beekeeper names, hive IDs, lab reports and sensor streams are synthetic (no real individuals); real pilot data replaces them in Phase 0.
- **AI data path:** prototype model runs on simulated telemetry; production CNN to be trained on open beehive datasets (refs 8, 9) plus KVIC apiary recordings.

Research takeaways applied in the design

- Engineered syrups can pass C4/HMF but fail NMR → ledger is lab-agnostic and anchors NMR + SMR results (refs 1, 4, 5)
- Hive vibration/acoustics can forewarn swarming (ref 7) → colony-sound feature in the model; CNN trained on open datasets (refs 8, 9) is the production path
- Blockchain honey frameworks exist (refs 10, 11); Honey Chain focuses on NMR/SMR-aware labs, Merkle-anchored telemetry and an offline bilingual PWA for Indian clusters
- Prototype lab limits are indicative (FSSAI / BIS / Codex style); final limits come from the notified lab

Technology and research

7. Ramsey et al. – "The prediction of swarming in honeybee colonies using vibrational spectra", *Sci. Rep.* **10**, 9798 (2020)
nature.com/articles/s41598-020-66115-5
8. Kulyukin – "Audio, image, video, and weather datasets for continuous electronic beehive monitoring", *Applied Sciences* **11**(10), 4632 (2021)
[Open datasets for training the production CNN](#)
9. **Machine Learning and Computer Vision Techniques in Continuous Beehive Monitoring Applications: A survey**
arxiv.org/pdf/2208.00085
10. "Fujairah Honey Chain (FHC): A Blockchain Framework for Monitoring Honey Production", *Information* **16**(8), 626 (2025)
mdpi.com/2078-2489/16/8/626
11. "Analysing blockchain adoption in beekeeping: application of theoretical models", *Frontiers in Sustainable Food Systems* (2025)
frontiersin.org/journals/sustainable-food-systems/articles/10.3389/fsufs.2025.1566341/full

TRY THE PROTOTYPE · WATCH THE DEMO



Demo video (3 min)
youtu.be/-nWbJNHHzBw



Live app
honey-chain-rust.vercel.app



Source code
Suyash2527/HoneyChain



Verify a sample
batch HC-2026-0008